A Project report on

Intracranial Hemorrhage detection

A Dissertation submitted to JNTU Hyderabad in partial fulfillment of the academic requirements for the award of the degree.

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

K.SAI PUNEETH
(20H51A0596)

B.VISHWANI
(20H51A05B3)

U. ANWESH GOUD
(20H51A05J8)

Under the esteemed guidance of

Mr. M. SHIVA KUMAR
(**Assistant Professor**)

Department of Computer Science and Engineering

CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC Autonomous)
*Approved by AICTE *Affiliated to JNTUH *NAAC Accredited with A^+ Grade

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD - 501401.

2020- 2024

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD – 501401

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Major Project Phase I report entitled **"Intracranial Hemorrhage detection "** being submitted by K.Sai Puneeth (20H51A0596), B.Vishwani (20H51A05B3), U.Anwesh Goud (20H51A05J8) in partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering** is a record of bonafide work carried out his/her under my guidance and supervision.

The results embodies in this project report have not been submitted to any other University or Institute for the award of any Degree.

Mr.M.Shiva Kumar
Assistant Professor
Dept. of CSE

Dr. Siva Skandha Sanagala
Associate Professor and HOD
Dept. of CSE

ACKNOWLEDGEMENT

With great pleasure we want to take this opportunity to express my heartfelt gratitude to all the people who helped in making this project work a grand success.

We are grateful to **Mr.M.Shiva Kumar, Assistant Professor** , Department of Computer Science and Engineering for his valuable technical suggestions and guidance during the execution of this project work.

We would like to thank **Dr. Siva Skandha Sanagala,** Head of the Department of Computer Science and Engineering, CMR College of Engineering and Technology, who is the major driving forces to complete my project work successfully.

We are very grateful to **Dr. Vijay Kumar Koppula**, Dean-Academics, CMR College of Engineering and Technology, for his constant support and motivation in carrying out the project work successfully.

We are highly indebted to **Major Dr. V A Narayana,** Principal, CMR College of Engineering and Technology, for giving permission to carry out this project in a successful and fruitful way.

We would like to thank the **Teaching & Non- teaching** staff of Department of Computer Science and Engineering for their co-operation

We express our sincere thanks to **Shri. Ch. Gopal Reddy**, Secretary, CMR Group of Institutions, for his continuous care.

Finally, We extend thanks to our parents who stood behind us at different stages of this Project. We sincerely acknowledge and thank all those who gave support directly and indirectly in completion of this project work.

K.Sai Puneeth	20H51A0596
B.Vishwani	20H51A05B3
U.Anwesh goud	20H51A05J8

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	LIST OF FIGURES	ii
	ABSTRACT	iii
1	**INTRODUCTION**	1-3
	1.1 Problem Statement	2
	1.2 Research Objective	2
	1.3 Project Scope and Limitations	3
2	**BACKGROUND WORK**	4-12
	2.1. ICH Detection using double branch CNN	6
	2.1.1. Introduction	6
	2.1.2. Merits,Demerits and Challenges	6-7
	2.1.3. Implementation of ICH Detection using CNN	7-8
	2.2. ICH detection using Parallel Deep Convolutional Models	9
	2.2.1. Introduction	9
	2.2.2. Merits,Demerits and Challenges	9-10
	2.2.3. Implementation of ICH Detection using DCM	11-12
3	**RESULTS AND DISCUSSION**	**13**
	3.1. Performance metrics	14
4	**CONCLUSION**	15
	4.1 Conclusion	16
	REFERENCES	**18**

List of Figures

FIGURE NO.	TITLE	PAGE NO.
1	Proposed framework	8
2	Systematic view of the proposed methodology	11
3	Representation of subtypes of ICH	12
4	Proposed solution's performance	12

ABSTRACT

Brain Hemorrhage is the eruption of the brain arteries due to high blood pressure or blood clotting that could cause a traumatic injury or death. It is the medical emergency in which a doctor also need years of experience to immediately diagnose the region of the internal bleeding before starting the treatment. In this, the deep learning algorithms, Yolo model is proposed for the Brain Hemorrhage classification. We use head CT scan images dataset to boost the accuracy rate and computational power of the deep learning models. The major aim of this study is to use the abstraction power of deep learning on a set of fewer images because in most crucial cases extensive datasets are not available on the spot. We propose the image augmentation and im-balancing the dataset methods with Yolo model to design a unique architecture and named as Brain Hemorrhage Classification based on Neural Network. The performance of the proposed approach will be analysed in terms of accuracy, precision, sensitivity, specificity and F1-score.

CHAPTER 1
INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1.Problem Statement

Intracranial hemorrhage (ICH) occurs when a diseased blood vessel within the brain bursts, allowing blood to leak inside the brain. Several factors, like disruption of the blood-brain barrier and leakage of fluids and proteins, inflammatory cascade or sudden increase in intracranial pressure after bleeding, develop into a brain injury in the areas surrounding the blood spill.Rapid diagnosis of Intracranial hemorrhage is crucial to administer appropriate medical intervention, potentially saving lives and reducing morbidity. Traditional diagnostic methods rely on radiologists manually interpreting Computed Tomography (CT) scans, a process that can be time-consuming and subject to human error.

1.2 Research Objective

To evaluate and compare the efficacy and efficiency of deep learning architectures, specifically YOLO (You Only Look Once) and RCNN (Region-based Convolutional Neural Networks), in the automated detection of intracranial hemorrhages from medical imaging data. The research aims to identify which model provides higher accuracy, fewer false positives and negatives, faster processing times, and greater clinical relevance, with the ultimate goal of improving early diagnosis and patient outcomes in cases of intracranial hemorrhage. The core objective of this research is to evaluate and contrast two prominent deep learning architectures, namely YOLO (You Only Look Once) and RCNN (Region-based Convolutional Neural Networks), for their capability in automated intracranial hemorrhage detection from imaging modalities such as CT or MRI scans. By embarking on this comparative study, the research aspires not only to ascertain which of these models yields superior accuracy and fewer diagnostic errors, but also to determine which can process images more rapidly—a critical factor in emergency medical scenarios.

1.3 .Project Scope and Limitations

- Detect and classify intracranial hemorrhages in medical imaging, such as CT scans, using deep learning models like YOLO (You Only Look Once) and RCNN (Region-based Convolutional Neural Networks).
- Gather a dataset of CT scans that are labeled for the presence, location, and type of intracranial hemorrhages. This dataset will be used for training and testing the models.
- Enhance the quality of images, remove noise, standardize image sizes, and augment the dataset to increase variability and robustness.
- Use the preprocessed dataset to train YOLO and RCNN models for hemorrhage detection. Fine-tune the architectures for optimal performance on the task.
- Split the dataset into training, validation, and test sets. Use the validation set to tune hyperparameters and the test set to evaluate the model's generalization capabilities.

Limitations:

- Data Quality and Quantity: The accuracy of deep learning models largely depends on the quality and quantity of the data. Limited or biased datasets can result in overfitting or poor generalization.
- Computational Resources: Training deep learning models, especially on large datasets, requires significant computational power and memory. Not all institutions may have access to the required resources.
- Interpretability: Deep learning models are often considered as "black boxes". Understanding why a particular decision was made by the model can be challenging, which might be concerning in medical applications.
- Variability in Scans: Different CT scanners and settings can produce images with slight differences, which might affect the model's performance.
- Generalizability: The model's performance may vary across different patient populations, age groups, or other demographic factors.

CHAPTER 2
BACKGROUND WORK

CHAPTER 2
BACKGROUND WORK

Intracranial hemorrhage (ICH) is a potentially life-threatening condition where there is bleeding within the brain. Rapid detection and intervention are vital to increase the chances of a positive outcome for patients. Deep learning methods have gained popularity in medical imaging due to their ability to process and recognize complex patterns in large datasets.

YOLO (You Only Look Once):

YOLO is a real-time object detection system that looks at the entire image in a single pass. It divides the image into a grid, and each grid cell predicts a certain number of bounding boxes and class probabilities. For ICH detection:

- The CT images can be divided into grids.
- Each grid cell can predict the likelihood of a hemorrhage being present in that part of the image and the bounding box surrounding it.

The main advantage is speed, making it suitable for real-time or near-real-time detection.

RCNN (Region-based Convolutional Neural Networks):

RCNNs are a family of methods designed to detect objects within an image by proposing regions and then classifying them. The basic steps are:
Use a region proposal algorithm to identify potential bounding boxes in the image.
Use a CNN to extract features from each proposed region.
Classify each region using the extracted features.

Existing Methods

2.1 ICH Detection using double branch CNN

2.1.1. Introduction

Intracranial hemorrhage (ICH) relates to bleeding occurring within the intracranial vault. Regardless of the actual cause, a hemorrhage constitutes a major threat. Therefore, an accurate and rapid diagnosis is crucial for the treatment process and its success. ICH diagnosis relies on patient medical history, physical examination, and non-contrast computed tomography (CT) examination of the brain region. CT examination enables bleeding localization and can indicate the primary causes of ICH .There are several challenges related to the ICH diagnosis and treatment: the urgency of the procedure, a complex and time-consuming decision-making process, an insufficient level of experience in the case of novice radiologists, and the fact that most emergencies occur at nighttime. Thus, there is a significant need for a computer-aided diagnosis tool to assist the specialist. Nevertheless, the accuracy of automated hemorrhage detection should be sufficiently high for medical purposes.

2.1.2.Merits,Demerits and Challenges

Merits :

- Multi-task Learning: A double branch CNN can be designed to perform multiple tasks. For example, one branch could detect the presence of a hemorrhage while the other determines its type.
- Different Modalities: Different branches can be optimized to process different imaging modalities, thereby providing a comprehensive analysis.
- Rich Feature Extraction: With two branches, the network can be designed to extract both local and global features, or spatial and contextual features, improving the detection's accuracy.
- Reduction in False Positives: By considering different features or tasks in separate branches, the model might be better equipped to reduce false positives.

Demerits:

- Increased Complexity: Managing a double branch structure can increase the complexity of the network, making it harder to train and optimize.
- Computational Costs: Training a double branch CNN might require more computational resources than a single branch network.
- Overfitting Risks: Due to the additional parameters introduced by the double branch structure, there might be an increased risk of overfitting, especially with smaller datasets.

Challenges:

Data Requirement: Training such a complex structure typically requires a large amount of labeled data. In medical imaging, obtaining a sufficient quantity of annotated data can be challenging.

Synchronization: Ensuring that the two branches are in sync and contribute effectively to the final decision can be challenging. It requires careful design and optimization.

Interpretability: Double branch CNNs, like other deep learning models, can act as black boxes, making it hard to interpret their decision-making process. In medical applications, interpretability is crucial for clinical acceptance.

Integration with Clinical Workflows: Designing a system that fits seamlessly into existing clinical workflows without causing disruptions is always a challenge.

Generalization: Ensuring that the model generalizes well across different devices, patient populations, and institutions is crucial.

2.1.3.Implementation of ICH Detection using double branch CNN :

The implementation of Intracranial Hemorrhage (ICH) detection using a double branch CNN requires designing a deep learning model that can process CT scan images (or other imaging modalities) and detect the presence of hemorrhages. The double branch approach can be leveraged to extract different types of features or handle different tasks.

It involves the design and training of a neural architecture that processes medical images through two parallel pathways to extract complementary features or perform simultaneous tasks. The two branches might be tailored to handle different image modalities, capture both local and global image features, or perform separate tasks such as detection and classification of the hemorrhage. Once the input image passes through both branches, the extracted features are merged, often through concatenation or fusion techniques. This combined feature set is then processed by subsequent layers for the final classification or detection of ICH. The implementation necessitates a sufficiently large labeled dataset to train the network and ensure generalizability across various scenarios. Regularization techniques might be employed to mitigate overfitting due to the added complexity of the double branch structure. Upon successful training, the model can be integrated into medical diagnostic systems to aid in the rapid and accurate detection of ICH.

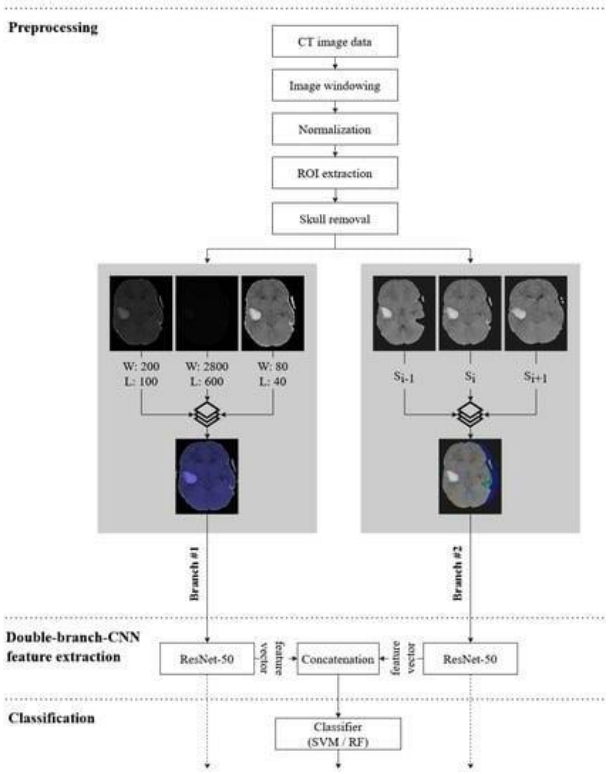

Figure 1 : Proposed framework

2.2. ICH detection using Parallel Deep Convolutional Models

2.2.1. Introduction

Intracranial Hemorrhage (ICH) is a critical medical condition characterized by bleeding within the brain tissues or cavities. Early and accurate detection of ICH is of paramount importance, as prompt treatment can significantly influence patient outcomes. However, as the complexity of the problem and diversity of the data increase, there is a growing need for more sophisticated architectures. One such advancement is the use of parallel convolution models. In this setup, multiple CNN branches operate simultaneously on the input image, often with different kernel sizes or structures. Parallel convolution models and boosting mechanisms offer a potent approach to ICH detection. The parallel convolutions ensure diverse and rich feature extraction, while the boosting mechanism ensures model robustness and improved accuracy by giving higher weight to challenging cases.

2.2.2. Merits,Demerits and Challenges

Merits:

- Comprehensive Feature Extraction: Parallel convolution models can extract features at various scales or orientations, making them effective for detecting anomalies that might be missed by a single convolution pathway.
- Accuracy Improvement: Boosting mechanisms focus on improving the performance iteratively by concentrating on the errors, which can lead to a more accurate model.
- Robustness: The ensemble nature of boosting can increase the robustness of the detection system as it relies on multiple models.
- Error Reduction: As boosting gives more weight to previously misclassified examples, it ensures that the algorithm becomes more sensitive to challenging cases, potentially reducing false negatives and false positives.
- Flexibility: Parallel convolution pathways can be tailored for specific features or types of hemorrhages, offering flexibility in model design.

Demerits:

- Increased Computational Cost: Both parallel convolutions and boosting require more computational power and memory due to the increased complexity.
- Training Time: With multiple pathways and iterative training for boosting, the training time is generally longer.
- Risk of Overfitting: While boosting can improve model accuracy, if not managed correctly, it can also lead to overfitting, especially if the number of boosting rounds is too high.
- Complexity: The integration of parallel convolutions with boosting can lead to a complex model, making it more challenging to understand, optimize, and maintain.

Challenges:

Data Requirement: Such complex models require substantial labeled data for training, which can be a challenge in the medical domain where annotated data is often scarce.

Interpretability: Deep learning models, especially complex ones like parallel convolution networks combined with boosting, can act as black boxes. In the medical field, interpretability is often a significant concern for clinicians and regulatory bodies.

Optimal Architecture Design: Determining the right number of parallel pathways, their configuration, and how many rounds of boosting to employ can be challenging and might require extensive experimentation.

Ensuring Generalizability: The model should generalize well across different devices, patient populations, and medical institutions. Given the complexity, there might be concerns about how well the model will perform in diverse scenarios.

2.2.3. Implementation of ICH detection using Parallel Deep Convolutional Models

Implementation of Intracranial Hemorrhage (ICH) detection using parallel convolution models and a boosting mechanism involves the simultaneous processing of medical images through multiple convolutional pathways to capture diverse features at various scales or resolutions. Each parallel convolution model extracts distinct features from the input image, ensuring comprehensive representation. After feature extraction, a boosting mechanism is employed to sequentially train a series of weak classifiers, where each subsequent classifier focuses on the errors made by its predecessor. This iterative process aims to improve the model's accuracy by giving more emphasis to misclassified instances. The final decision on ICH presence or classification is made by aggregating the outputs of these weak classifiers, often through a weighted majority vote. The combined strength of parallel convolutions, which provide diverse feature representations, and the boosting mechanism, which continually refines the decision boundary, offers a robust approach for accurate and reliable detection of ICH in medical imaging scenarios.

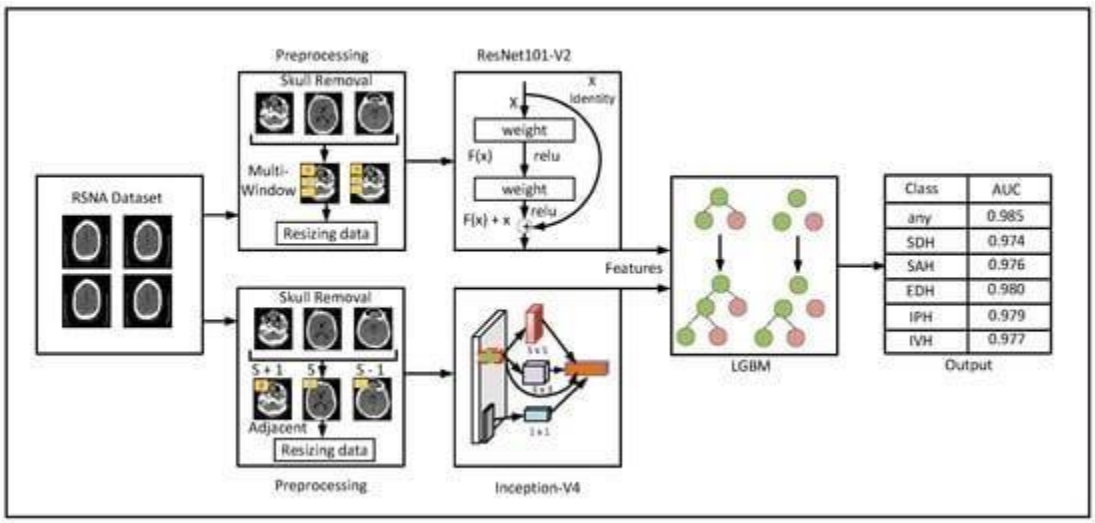

Figure 2 : Systematic view of the proposed methodology

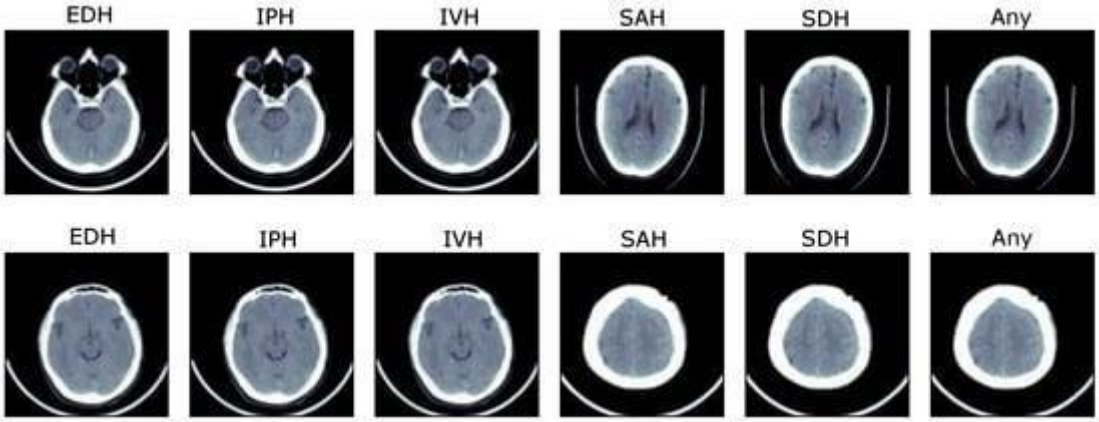

Figure 3 : Representation of subtypes of ICH

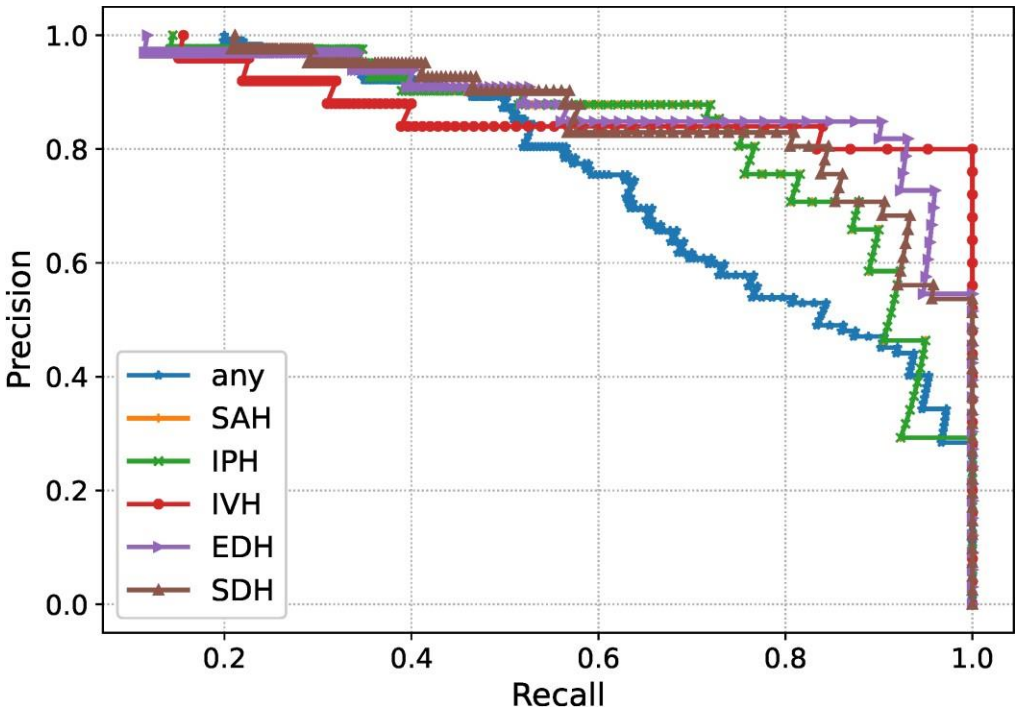

Figure 4: Proposed solution's performance

CHAPTER 3
RESULTS AND DISCUSSION

CHAPTER 3
RESULTS AND DISCUSSION

3.1 Performance metrics

Performance metrics are crucial in assessing the effectiveness of deep learning models, especially in critical applications like medical imaging. When evaluating the performance of deep learning models like RCNN and YOLO for intracranial hemorrhage (ICH) detection, several metrics are pivotal. Accuracy measures the overall correct classifications, while Sensitivity (or Recall) and Specificity assess the model's effectiveness in identifying true positives and true negatives, respectively. Precision provides insight into the proportion of predicted hemorrhages that are actual cases, and the F1-Score offers a balance between precision and recall. For these object detection frameworks, the Mean Average Precision (mAP) is essential, indicating the model's precision across various types and locations of hemorrhages. Intersection over Union (IoU) evaluates the accuracy of the model's localization by measuring the overlap between predicted and actual bounding boxes. Additionally, false positives and false negatives are crucial for understanding model errors, especially in the critical context of medical diagnostics. Since RCNN and YOLO have different architectural complexities, metrics such as frames per second (FPS) during inference, model size, and training time can be useful to understand their practical implications in real-world, time-sensitive applications.

CHAPTER 4
CONCLUSION

CHAPTER 4
CONCLUSION

In conclusion, the utilization of deep learning methodologies, notably the RCNN and YOLO models, for the detection of Intracranial Hemorrhage (ICH) represents a significant advancement in medical imaging analysis. The RCNN, with its region-proposal mechanism, offers a precise means to detect and classify hemorrhages, prioritizing accuracy. In contrast, YOLO, with its end-to-end single-pass architecture, emphasizes speed without substantially compromising detection performance. Both models, through their unique architectures and strengths, facilitate rapid and accurate ICH detection, crucial for timely medical interventions. By integrating these advanced deep learning techniques into the clinical workflow, the potential exists not only to augment the capabilities of radiologists but also to save lives by ensuring swift, informed decisions in critical scenarios. However, the practical implementation of these models necessitates rigorous validation, continuous updates based on the latest data, and careful consideration of ethical implications associated with automated medical diagnoses.

REFERENCES

REFERENCES

[1]. Arbabshirani MR, Fornwalt BK, Mongelluzzo GJ, Suever JD, Geise BD, Patel AA et al. Advanced machine learning in action: identification of intracranial haemorrhage on computed tomography scans of the head with clinical workflow integration. NPJ Digit Med. 2018. Apr 4;1:9. doi: 10.1038/s41746-017-0015-z ; PMCID: PMC6550144. https://pubmed.ncbi.nlm.nih.gov/31304294/ [PMC free article] [PubMed] [CrossRef] [Google Scholar]

[2]. Ojeda P, Zawaideh M, Mossa-Basha M, Haynor D. The utility of deep learning: evaluation of a convolutional neural network for detection of intracranial bleeds on non-contrast head computed tomography studies. Proc. SPIE 10949, Medical Imaging. 2019; Image Processing, 109493J. 10.1117/12.2513167 [CrossRef] [Google Scholar]

[3]. Chilamkurthy, S. et al. Deep learning algorithms for detection of critical findings in head CT scans: A retrospective study. Lancet 392, 2388–2396. https://doi.org/10.1016/s0140-6736(18)31645-3 (2018).

[4]. https://www.mdpi.com/2075-4418/13/4/652

[5]. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0260560

[6]. https://www.mdpi.com/2076-3417/10/21/7577